Explainable AI Clinical Decision Support System

Clinical Prediction Report

Condition: Diabetes

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Patient Information

Field	Value
Patient ID	PAT003
Patient Name	Final Test Patient
Doctor / Clinician	Dr. Smith
Hospital / Clinic	Fortis Hospital
Disease	Diabetes

Clinical Parameters

Feature	Observed Value
Age	65
Gender	Male
Polyuria	Yes
Polydipsia	Yes
Sudden Weight Loss	Yes
Weakness	Yes
Polyphagia	Yes
Genital Thrush	Yes
Visual Blurring	Yes
Itching	Yes
Irritability	Yes
Delayed Healing	Yes
Partial Paresis	Yes
Muscle Stiffness	Yes
Alopecia	Yes
Obesity	Yes

Model Assessment

Field	Value
Condition	Diabetes
Prediction	Diabetes
Class	Diabetes
Probability	100.00%
Risk Level	High
Clinical Recommendation	High likelihood of diabetes. Consultation with a healthcare professional is strongly recommended for further clinical evaluation and appropriate diagnostic testing.

SHAP Evidence

Feature contribution to the model decision.

Feature	Contribution
Polyuria	0.1889
Polydipsia	0.0877
Irritability	0.0424
Age	0.0309
Partial Paresis	0.0265

LIME Evidence

Local explanation for this patient.

Feature / Rule	Weight
-0.99 < Polyuria <= 1.01	0.4549
-0.94 < delayed healing <= 1.06	0.1466
Obesity > -0.46	-0.1316
-0.77 < muscle stiffness <= 1.30	-0.1291
-0.87 < partial paresis <= 1.15	0.0512

SHAP-LIME Agreement

Consistency between the SHAP and LIME local explanation methods.

Field	Value
Agreement Score	68.00%
Feature Overlap	20.00%

Field	Value
Direction Agreement	100.00%
Ranking Agreement	100.00%
Overall Consistency	68.00%
Consistency Level	Moderate
Shared Feature Count	1
Shared / Common Features	Polyuria
SHAP Only	Polydipsia, Irritability, Age, Partial Paresis
LIME Only	Delayed, Obesity, Muscle, Partial

Agreement measures consistency between SHAP and LIME explanations. It indicates explanation consistency, not clinical certainty.

Multi-Feature Counterfactual Explanation

Counterfactual explanations identify hypothetical alternative patient profiles in which one or more clinical features are changed to determine whether the model prediction can change.

Field	Value
Original Prediction	Diabetes
Original Probability	100.00%

Counterfactual #1

Field	Value
New Prediction	No Diabetes
New Probability	0.00%
Features Changed	2
Change Magnitude	2.00
Changed Clinical Features	Gender: 1 → 0 Polyuria: 1 → 0

Counterfactual #2

Field	Value
New Prediction	No Diabetes
New Probability	0.00%
Features Changed	2
Change Magnitude	2.00
Changed Clinical Features	Delayed Healing: 1 → 0 Polyuria: 1 → 0

Counterfactual #3

Field	Value
New Prediction	No Diabetes
New Probability	0.00%
Features Changed	2
Change Magnitude	2.00
Changed Clinical Features	Obesity: 1 → 0 Polyuria: 1 → 0

Counterfactual #4

Field	Value
New Prediction	No Diabetes
New Probability	0.00%
Features Changed	2
Change Magnitude	2.00
Changed Clinical Features	Partial Paresis: 1 → 0 Polydipsia: 1 → 0

Counterfactual #5

Field	Value
New Prediction	No Diabetes
New Probability	0.00%
Features Changed	2
Change Magnitude	6.00
Changed Clinical Features	Age: 65 → 70 Sudden Weight Loss: 1 → 0

What does this mean?

Counterfactual explanations show hypothetical patient profiles in which one or more model input features are changed and the resulting prediction is evaluated. These results illustrate the model's decision boundary.

Important: Counterfactual scenarios are model-generated hypothetical examples. They must not be interpreted as clinical treatment targets, prescriptions, diagnoses, or instructions to change a patient's health measurements.

Clinical Interpretation Note

The prediction and explanation values in this report represent the output of the machine-learning system for the submitted patient data. SHAP and LIME provide model interpretability, while counterfactual analysis illustrates hypothetical changes that may cross the model's decision boundary.

Disclaimer: This report is intended for clinical decision support only. Model outputs, SHAP values, LIME explanations, agreement scores, and counterfactual scenarios must not be interpreted as definitive diagnoses, prescriptions, treatment instructions, or required clinical targets. Final clinical decisions must be made by qualified healthcare professionals using appropriate clinical assessment and additional evidence.

Generated by Explainable AI Clinical Decision Support System

Doctor Signature

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Dr. Smith

Patient Signature

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Final Test Patient